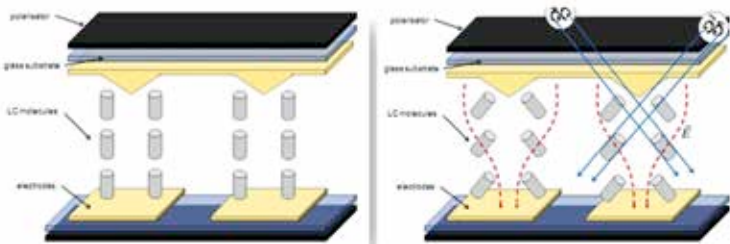


observe fast moving objects or if you happen to be playing a fast-paced game, then the ghosting effect becomes a lot more prominent and that makes the display unattractive for gamers.

As the panel technology evolved, we saw improvements in contrast ratios, especially in P-MVA and S-MVA panels. Current generation of Advanced-MVA (A-MVA) panels offer way better response times. To the extent that Sharp, a prominent display manufacturer, is planning on releasing a VA panel with a refresh rate of 120 Hz. However, despite all these improvements, it is unlikely that you'll come across Ultra-wide or curved A-MVA panels. Most monitors sporting A-MVA panels now offer 5-6 ms G2G response times with overdrive technologies pulling it down much lower. And this has led to an increased level of adoption among gamers.

PVA

In PVA panels, the crystals have the same structure as those present in MVA, i.e. domains with different orientations of crystals. Viewing angles aren't exactly the same as MVA and there is a slight difference in the way



Patterned Vertical Alignment technology isn't that different from MVA

colours appear when you are looking straight at the screen. However, current generation of PVA panels don't experience such a pronounced 'off-centre' contrast shift or 'black crush' as it came to be known.

PVA panels ended up facing the same problems as MVA – poor response times, black crush, excessive blurring. These problems were tackled as the technology evolved and more panel subtypes were developed. Response times, in particular, were improved upon with the advent of overdrive technologies which essentially applies more voltage to make the crystals to shift faster from one orientation to another.